Stefan M Arseneau

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My research focuses on precision measurements of white dwarf stellar parameters and their use in stellar structure inference. I am especially interested in efficient uncertainty characterization with MCMC, statistical inference using large and noisy datasets, and identifying previously unknown sources of systematic uncertainty.

Education

Boston University — PhD, MA in Astronomy — Expected 2029

Johns Hopkins University — BS in Physics, BA in Mathematics — 2024

Experience

Research Fellow, Department of Astronomy, Boston University — Boston, MA, USA — 2024 — Present

- Performed observational measurements of the white dwarf mass-radius relation and stellar structure using gravitational redshifts and asteroseismology.

Member, Milky Way Mapper, Sloan Digital Sky Survey V — 2024 — Present

- Developed and validated white dwarf parameter estimation for `astra`, Milky Way Mapper data reduction pipeline.

Research Fellow, CLASS Telescope, Johns Hopkins University — Baltimore, MD, USA — 2022 — 2024

- Modeled effects of radiative heat transfer on performance of filters using COMSOL Multiphysics.

Talks & Conferences

- Conference Organizer — Boston Area Planetary Science Meeting — Spring 2025 & Fall 2025
- Poster Presentation — SDSS-V Collaboration Meeting (MPIA, Heidelberg, DE) — 2025

  Resolution-Corrected White Dwarf Gravitational Redshifts Validate SDSS-V Wavelength Calibration and Enable Accurate Mass-Radius Tests

Observatory Allocations

- ESO Very Large Telescope, UVES, 17.5 Hours (co-PI with Roberto Raddi) — 2024

  Constraints on the Gravitational Redshifts and Mass-Radius Relation of High-Mass White Dwarfs in Wide Binaries
- Gemini Observatory, GMOS, 2 Hours — 2023

  Probing the Mass-Radius Relation of White Dwarfs in Wide Binaries

Honors

- Graduate Research Fellowship, Boston University — 2025
- Provost's Undergraduate Research Award, Johns Hopkins University — 2023

Open Source Projects

Compact Object Radial Velocities — github.com/vedantchandra/corv

- Primary author and maintainer of a scientific software package for precision radial velocity estimation.
- Integrated into the Sloan Digital Sky Survey data reduction pipeline, processing >200 GB per night.

Publications

Publications as First Author —

3. **S. M. Arseneau**; J.J. Hermes; M.E. Camisassa; R. Raddi; E.B. Bauer, 2026, *Constraints on White Dwarf Hydrogen Layer Masses Using Gravitational Redshifts*, The Astrophysical Journal, 1000 297 (arXiv:2603.02314) [ADS Link].

2. **S. M. Arseneau**; J.J. Hermes; N.L. Zakamska; K. El-Badry; N.R. Crumpler; V. Chandra; G. Adamane Pallathadka; C. Badenes; B.T. Gäniscke; N. Gentile Fusillo, 2025, *Resolution-Corrected White Dwarf Gravitational Redshifts Validate SDSS-V Wavelength Calibration and Enable Accurate Mass-Radius Tests*, The Astrophysical Journal, 991 190 (arXiv:2508.04775) [ADS Link]

1. **S. M. Arseneau**; V. Chandra; H.C. Hwang; N.L. Zakamska; G.A. Pallathadka; N.R. Crumpler; J.J. Hermes; K. El-Badry; H-W. Rix; K.G. Stassun; B.T. Gänsicke; J.R. Brownstein; S. Morrison, 2025, *Measuring the Mass–Radius Relation of White Dwarfs Using Wide Binaries*, The Astrophysical Journal, 963, 17 (arXiv:2310.19866) [ADS Link]

Publications as Contributing Author —

10. G.A. Pallathadka; V. Chandra; N.L. Zakamska; N.R. Crumpler; **S. M. Arseneau**; K. El-Badry; B.T. Gäniscke; Y. Zentai; J. J. Hermes; A.D. Schwope; C. Badenes; N.P. Gentile Fusillo; S. Morrison; T. Cunningham; P. Chakraborty; G Tovmasian; D Bizyaev; K. Pan; S.F. Anderson; S. Demasi, 2025, *Double White Dwarf Binaries in SDSS-V DR19 : A catalog of DA white dwarf binaries and constraints on the binary population*, The Astrophysical Journal, In Press (arXiv:2509.02906) [ADS Link]

9. K.R. Helson; C.Y.Y. Chan; **S. Arseneau**; A. Barlis; C.L. Bennett; T.M. Essinger-Hileman; H. Guo; T. Marriage; M.A. Quijada; A.E. Tokarz; S.L. Vivod; E.J. Wollack, 2025, *Diamond-loaded polyimide aerogel scattering filters and their applications in astrophysical and planetary science observations,* Review of Scientific Instruments, Submitted (arXiv:2508.20406) [ADS Link]

8. N.R. Crumpler; V. Chandra; N.L. Zakamska; G.A. Pallathadka; **S. Arseneau**; N.P. Gentile Fusillo; J. J. Hermes; C. Badenes; P. Chakraborty; B.T. Gänsicke; S. Morrison; H-W. Rix; S.P. Schmidt; A. Schwope; K.G. Stassun, 2025, *A Large Catalog of DA White Dwarf Characteristics Using SDSS and Gaia Observations,* The Astrophysical Journal, 989 24 (arXiv:2508.00818) [ADS Link]

7. G.A. Pallathadka; V. Chandra; B.T. Gänsicke; N.L. Zakamska; D. Koester; Y. Zentai; N.R. Crumpler; **S. M. Arseneau**; J.J. Hermes; M.R. Schreiber; K.G. Stassun; A. Schwope; K. El-Badry; G. Tovmassian; T. Cunningham; S. Morrison, 2025, *Double White Dwarf Binaries in SDSS-V DR19 : The discovery of a rare DA+DQ white dwarf binary with 31 hour orbital period,* The Astrophysical Journal, In Press (arXiv:2507.11618) [ADS Link]

6. J.A. Kollmeier; H.-W. Rix; C. Aerts; J. Aird; P.V. Alfaro; A. Almeida; S.F. Anderson; O.J. Arranz; **S. M. Arseneau** et al., 2025, *Sloan Digital Sky Survey-V: Pioneering Panoptic Spectroscopy,* The Astronomical Journal, 171, 52 (arXiv:2507.06989) [ADS Link]

5. SDSS Collaboration, G.A. Pallathadka, M. Aghakhanloo, J. Aird, A. Almeida, S. Amrita, F. Anders, S.F. Anderson, **S. Arseneau** et al., 2025, *The Nineteenth Data Release of the Sloan Digital Sky Survey,* The Astrophysical Journal, In Press (arXiv:2507.07093) [ADS Link]

4. N.R. Crumpler; V. Chandra; N.L. Zakamska; G.A. Pallathadka; **S. Arseneau**; N. Gentile Fusillo; J.J. Hermes; C. Badenes; P. Chakraborty; B.T. Gänsicke; S.P. Schmidt, 2024, *Detection of the Temperature Dependence of the White Dwarf Mass-Radius Relation with Gravitational Redshifts,* The Astrophysical Journal, 977 2 (arXiv:2412.14331) [ADS Link]

3. G.A. Pallathadka; V. Chandra; N.L. Zakamska; H-C. Hwang; Y. Zentai; J.J. Hermes; K. El-Badry; B.T. Gänsicke; S. Morrison; N.R. Crumpler; **S. Arseneau**, 2024, *Discovery of A Proto-White Dwarf With A Massive Unseen Companion,* Astrophysical Journal, 968 1 (arXiv:2310.16313) [ADS Link]

2. A. Barlis; **S. Arseneau**; C.L. Bennett; T. Essinger-Hileman; H. Guo; K.R. Helson; T. Marriage; M.A. Quijada; A.E. Tokarz; S.L. Vivod; E.J.Wollack, 2022, *Characterization of aerogel scattering filters for astronomical telescopes,* Proc. SPIE 12190, Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI, 121902I (arXiv:2208.04257) [ADS Link]

1. K.R. Helson; **S. Arseneau**; A. Barlis; C.L. Bennett; T.M. Essinger-Hileman; H. Guo; T. Marriage; M.A. Quijada; A.E. Tokarz; S.L. Vivod; E.J. Wollack, 2022, *Novel infrared-blocking aerogel scattering filters and their applications in astrophysical and planetary science,* Proc. SPIE 12190, Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI, 121901P (arXiv:2208.03755) [ADS Link]

Non-Peer-Reviewed Publications —

2. G.A. Pallathadka et. al, incl. **S. M. Arseneau**; 2026. Constraints on the Double White Dwarf Binary Population from SDSS-V DR19. American Astronomical Society Meeting Abstracts, 247, 428.01. [ADS Link]

1. G.A. Pallathadka et. al, incl. **S. M. Arseneau**; 2025. Double Degenerate WD Binaries in SDSS-V DR19. American Astronomical Society Meeting Abstracts, 245, 257.04. [ADS Link]

Professional References

- J. J. Hermes (Boston University, PhD Advisor) jjhermes@bu.edu
- Nadia L. Zakamska (Johns Hopkins University, Advisor) zakamska@jhu.edu
- Carles Badenes (University of Pittsburgh) badenes@pitt.edu